Fiscal Year:	FY 2007	**Task Last Updated:**	FY 01/31/2008
PI Name:	Zderic, Vesna Ph.D.		
Project Title:	Contrast Agents for High-Intensity Focused Ultrasound Hemostasis		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Smart Medical Systems Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	98195	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2004 NSBRI-RFP-04-01 Postdoctoral Fellowships
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No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Crum, Lawrence (MENTOR/University of Washington)		
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POSTDOCTORAL FELLOWSHIP
(1) the original aims:

The hypothesis is that application of ultrasound contrast agents can improve efficacy of HIFU hemostasis of injured solid organs, while maintaining treatment safety.

Specific Aim 1: Determination of efficacy and safety of HIFU hemostasis in the presence of ultrasound contrast agents (UCA).

Aim 1a: Mechanisms of HIFU hemostasis in the presence of commercially available UCA Optison.

Task Description:	Aim 1b: Efficacy and safety of HIFU hemostasis in the presence of Optison. Specific Aim 2: Development and testing of autologous UCA for HIFU hemostasis. Aim 2a: Development of autologous UCA. Aim 2b: Determination of the efficacy of HIFU hemostasis in the presence of autologous UCA. (2) the key findings: Thermal measurements showed similar temperatures of the coagulum formed at the injury site treated with HIFU, with or without UCA administration. Inertial cavitation activity, a signature of mechanical effects, was higher during HIFU application in the presence of UCA. The presence of Optison in the blood stream during HIFU hemostasis of liver injuries resulted in 37% faster times ($p<0.05$) needed to achieve complete hemostasis as compared to control treatments (with no UCA present). The presence of in-house UCA in the blood stream also resulted in faster hemostasis times for sealing of liver injuries (46% faster as compared to control treatments, $p<0.05$). Coagulum formation at the injury site was faster as the concentration of administered UCA increased. Hemostatic seal over the injury site achieved with HIFU or HIFU+UCA was more robust as compared to the seal achieved with electrocautery. UCA (both Optison and in-house UCA) increased the rate of successful bleeding detection and localization of hidden solid organ injuries (in rabbit kidneys and liver) and vessel injuries (in rabbit and porcine arteries). With regular ultrasound imaging methods (B-mode and Color Doppler) injuries with slow bleeding could not be detected unless UCA were present in the blood stream. The levels of systemic blood hemolysis during HIFU therapy increased if UCA were administered before the therapy. Light microscopy observations showed no histological difference between the liver tissue treated with HIFU only or HIFU in the presence of UCA. The presence of in-house UCA resulted in 35% reduction of the lesion depth (i.e., resulted in the production of shallow lesions localized at the injury site), and prevented the formation of HIFU lesions beyond the focus. The presence of bubbles at the HIFU focus prevented the temperature increase at the tissue-air interface in the post-focal region. (3) the impact of these findings : Both in-house UCA and commercially available UCA Optison can be used to achieve faster hemostasis times of liver injuries and faster coagulum formation at the injury site. Our findings showed that faster coagulum formation and hemorrhage control was due to the mechanical effects of HIFU application in the presence of UCA. Bleeding detection experiments showed that UCA (both Optison and in-house UCA) can be used to improve the detection and localization of bleeding sites, which could be of significant importance during HIFU treatment of hidden bleedings and for eventual extracorporeal HIFU application.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	On Earth, bleeding due to blunt trauma is one of the primary causes of death within an hour after the injury if adequate care is not provided. Rapid achievement of hemostasis is necessary to prevent hemorrhagic shock following intra-abdominal injury. However, hemostasis is not an easy task in trauma patients, due to the usual presence of coagulopathy, resulting from hypothermia, acidosis, and dilution of clotting factors. Difficulties associated with treating battlefield and civilian trauma hemorrhage are in being able to detect and localize the hemorrhage site, and to stop the bleeding non-invasively, all within a very short time (of several minutes) after the trauma has occurred. Our group demonstrated potential of ultrasound-guided HIFU to both detect a bleeding site and deliver energy to produce hemostasis. Specifically, this NSBRI-funded project addressed HIFU-induced hemorrhage control from solid organs in the presence of ultrasound contrast agents (UCA). The administration of UCA may facilitate detection and localization of bleeding sites, and allow faster hemorrhage control. Our eventual goal is to develop a portable automatic HIFU device that can be used at the site of accident to localize and stop the bleeding by a non-skilled operator, similar to current usage of defibrillators.
Task Progress:	Specific tasks accomplished in the Project year 2 are described below: 1. Development of in-house ultrasound contrast agents (UCA) for High Intensity Focused Ultrasound (HIFU) therapy. The size of UCA had to be optimized to be less than 10 μm to allow passage through lungs. A variety of UCA manufacturing methods were explored. Our current in-house UCA solution consists of 5% albumin and 10% dextrose in saline. This solution is sonicated in the presence of perfluoropropane gas for 90 s using 20 kHz sonicator (at ~40 W/cm^2) to produce microbubbles of appropriate size. These in-house UCA have concentration of 3x10^8 bubbles/ml with the bubble size of 1-8 μm (mean size of 3 μm). 2. Bleeding detection in the presence of in-house UCA. Scalpel cuts (3 cm in length and 0.5 cm deep) were produced in the posterior surface of rabbit livers and kidneys; or puncture injuries were produced transcutaneously in the rabbit femoral arteries. The in-house UCA made of albumin, dextrose and perfluoropropane gas improved detection and localization of slow bleeding from solid organ injuries and punctured vessels. This bleeding is otherwise not detected using standard ultrasound methods. 3. Determination of the efficiency of HIFU treatment in the presence of in-house UCA. HIFU treatments (at the frequency of 5.5 MHz, in-situ intensity of 6,800 W/cm^2) of liver lacerations were performed in rabbit in-vivo, with or without the presence of in-house UCA (0.8 ml, 1:2 saline dilution) in the blood stream. The average hemostasis times

	normalized with the bleeding rate were 39 s/ml*min for HIFU only treatments and 21 s/ml*min for HIFU+UCA treatments. The presence of home-made UCA in the blood stream resulted in 44% percent reduction in the times to the first formation of the blood coagulum at the HIFU treatment site, and 46% reduction ($p<0.05$) in HIFU hemostasis times. 4. Determination of the size and shape of HIFU lesions in the rabbit liver in-vivo, which were produced immediately after the injection of in-house UCA or without the presence of UCA. The objective of this study was to see whether it would be possible to customize the shape of HIFU lesions such that the lesion is localized at the site of interest (i.e., organ surface injury), without causing damage in the healthy deep tissue regions. The presence of in-house UCA resulted in 35% reduction of the lesion depth (i.e., resulted in the production of more shallow lesions). The maximal diameter of the lesions increased by 30% in the presence of in-house UCA, as compared to control lesions. It appeared that the presence of in-house microbubbles in the liver tissue caused the shielding of the postfocal region from the HIFU exposure.
Bibliography Type:	Description: (Last Updated: 09/06/2019)
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